MA385 Part 3: Linear Algebra 1

# 3.3 LU-factorisation

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Tadeusz Banachiewicz (left), was polish mathematician who first presented Gaussian Elimination as matrix factorization/decomposition. Alan Turing (right) formulated LU-decomposition as we know it today and, moreover, gave the first analysis of the impact of round-off error. Photos: Wikipedia.

# 1. Outline of Section 3.3

*In these slides,*

- *LT means "lower triangular"*
- *UT means "upper triangular"*


1 A formula for LU-factorisation

2 Existence of an $LU$-factorisation

3 Exercises


For more, see Section 2.3 of Suli and Mayers:
https://ebookcentral.proquest.com/lib/nuig/reader.action?docID=221072&ppg=51&c=UERG

# 1. Outline of Section 3.3

The goal of this section is to demonstrate that the process of Gaussian Elimination applied to a matrix $A$ is equivalent to factoring $A$ as the product of a unit lower triangular and upper triangular matrix.

The Section 3.2 we saw that each elementary row operation in Gaussian Elimination involves replacing $A$ with $(I + \mu_{rs}E^{(rs)})A$.

**Example:** For the $3 \times 3$ case, this involved computing

$$(I + \mu_{32}E^{(32)})(I + \mu_{31}E^{(31)})(I + \mu_{21}E^{(21)})A.$$

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In general we multiply $A$ by a sequence of matrices

$$(I + \mu_{rs} E^{(rs)}),$$

all of which are **unit lower triangular** (=unit LT) matrices.
When we are finished we have reduced $A$ to an **upper triangular** (UT) matrix.
So we can write the whole process as

$$L_k L_{k-1} L_{k-2} \dots L_2 L_1 A = U, \tag{1}$$

where each of the $L_i$ is a unit LT matrix.

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However, we know from Section 3.2 that the product of **unit LT** matrices is itself a unit LT matrix. So we can write the whole process described in (1) as

$$\tilde{L}A = U. \tag{2}$$

Also from Section 3.2, the inverse of a **unit LT** matrix exists and is a **unit LT** matrix. So we can write (2) as

$$A = LU$$

where $L$ is unit lower triangular and $U$ is upper triangular.
This is called "$LU$-**factorisation**".

# 1. Outline of Section 3.3

**Definition 3.4.1**

The $LU$-**factorization** of the matrix is a unit lower triangular matrix $L$ and an upper triangular matrix $U$ such that $LU = A$.

**Example 3.4.1**

If $A = \begin{pmatrix} 3 & 2 \\ -1 & 2 \end{pmatrix}$ then:

# 1. Outline of Section 3.3

**Example 3.4.2**

If $A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}$ then:

# 1. Outline of Section 3.3

You should find

$$\underbrace{\begin{pmatrix} 3 & -1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & -4 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 0 & 3/7 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 3 & -1 & 1 \\ 0 & 14/3 & 7/3 \\ 0 & 0 & -5 \end{pmatrix}}_{U}.$$

# 2. A formula for LU-factorisation

We now want to work out formulae for $L$ and $U$ where

$$a_{ij} = (LU)_{ij} = \sum_{k=1}^{n} l_{ik} u_{kj} \qquad 1 \leq i, j \leq n.$$

Since $L$ and $U$ are triangular,

$$\text{If } i \leq j \quad \text{then} \quad a_{ij} = \sum_{k=1}^{i} l_{ik} u_{kj} \tag{3a}$$

$$\text{If } j < i \quad \text{then} \quad a_{ij} = \sum_{k=1}^{j} l_{ik} u_{kj} \tag{3b}$$

# 2. A formula for LU-factorisation

The first of these equations can be written as

$$a_{ij} = \sum_{k=1}^{i-1} l_{ik} u_{kj} + l_{ii} u_{ij}.$$

But $l_{ii} = 1$ so:

$$u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} u_{kj} \quad \begin{cases} i = 1, \ldots, j-1, \\ j = 2, \ldots, n. \end{cases} \tag{4a}$$

And from the second:

$$l_{ij} = \frac{1}{u_{jj}} \left( a_{ij} - \sum_{k=1}^{j-1} l_{ik} u_{kj} \right) \quad \begin{cases} i = 2, \ldots, n, \\ j = 1, \ldots, i-1. \end{cases} \tag{4b}$$

**Example 3.4.3**

Find the $LU$-factorisation of

$$A = \begin{pmatrix} -1 & 0 & 1 & 2 \\ -2 & -2 & 1 & 4 \\ -3 & -4 & -2 & 4 \\ -4 & -6 & -5 & 0 \end{pmatrix}$$

# 2. A formula for LU-factorisation

**Full details of the example**: First, using (4a) with $i = 1$ we have $u_{1j} = a_{1j}$:

$$U = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{pmatrix}.$$

Then (4b) with $j = 1$ we have $l_{i1} = a_{i1}/u_{11}$:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & l_{32} & 1 & 0 \\ 4 & l_{42} & l_{43} & 1 \end{pmatrix}.$$

Next (4a) with $i = 2$ we have $u_{2j} = a_{2j} - l_{21}u_{2j}$:

$$U = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{pmatrix},$$

# 2. A formula for LU-factorisation

then (4b) with $j = 2$ we have $l_{i2} = (a_{i2} - l_{i1}u_{12})/u_{22}$:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \\ 4 & 3 & l_{43} & 1 \end{pmatrix}$$

Etc....

# 3. Existence of an $LU$-factorisation

Not every matrix has an $LU$-factorisation. So we need to characterise the matrices that do.
To prove the next theorem we need the Cauchy-Binet Formula: $\det(AB) = \det(A)\det(B)$.

# 3. Existence of an $LU$-factorisation

**Theorem 3.4.1**

If $n \geq 2$ and $A \in \mathbb{R}^{n \times n}$ is such that every leading principal submatrix of $A$ is nonsingular for $1 \leq k < n$, then $A$ has an $LU$-factorisation.

**Proof** The proof is by induction. First, let's take $n = 2$, ...

# 3. Existence of an $LU$-factorisation

Now assume that the theorem holds for all matrices of order up to $(n-1)$.
Let $A$ be an $n$-matrix. Partition it by the last row and column.
Then, if $A$ has an LU-factorization, it is of the form is

$$\left(\begin{array}{c|c} A^{(n-1)} & \mathbf{b} \\ \hline \mathbf{c}^T & d \end{array}\right) = \left(\begin{array}{c|c} L^{(n-1)} & \mathbf{0} \\ \hline \mathbf{w}^T & d \end{array}\right) \left(\begin{array}{c|c} U^{(n-1)} & \mathbf{x} \\ \hline \mathbf{0}^T & z \end{array}\right).$$

Now we need to show that all of $L^{(n-1)}$, $U^{(n-1)}$, $\mathbf{w}$, $\mathbf{x}$ and $z$ exist.

# 3. Existence of an $LU$-factorisation

**Exercise 3.4.1**

Many textbooks and computing systems compute the factorisation $A = LDU$ where $L$ and $U$ are unit lower and *unit* upper triangular matrices respectively, and $D$ is a diagonal matrix. Show such a factorisation exists, providing that if $n \geq 2$ and $A \in \mathbb{R}^{n \times n}$, then every leading principal submatrix of $A$ is non-singular for $1 \leq k < n$.